

Oxidative stress occurs in cells when reactive oxygen species (ROS) such as H_2O_2 , superoxide (O_2^-), and hydroxyl radicals ($\cdot\text{OH}$) interact destructively with biological molecules. Several studies have indicated that histidine and certain peptides that contain histidine can decrease oxidative stress.

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Experiment 1

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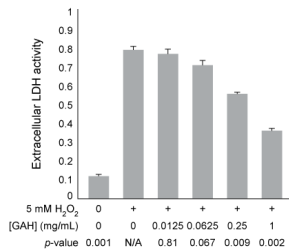


Figure 1 Extracellular LDH activity from PC12 cells exposed to H_2O_2 and varying concentrations of GAH. *p*-values are relative to 5 mM H_2O_2 and 0 mg/mL GAH.

Experiment 2

LDH was purified and assayed for activity in the presence of 5 mM H_2O_2 and varying concentrations of GAH (Figure 2).

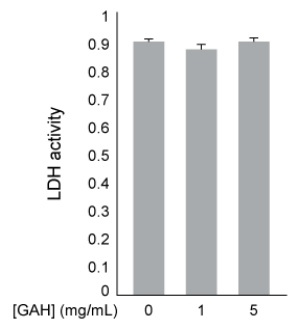


Figure 2 Activity of purified LDH with varying concentrations of GAH

Experiment 3

PC12 cells were incubated with or without 5 mM H_2O_2 in the presence and absence of 1 mg/mL GAH. Cells were then stained with the fluorescent marker TUNEL, which binds strongly to apoptotic cells. The percentage of apoptotic cells in each condition is shown in Figure 3.

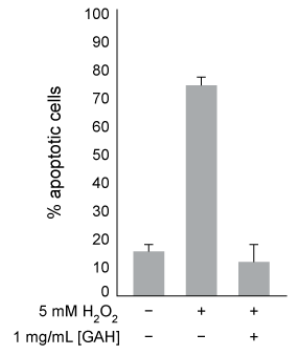
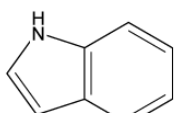
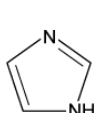
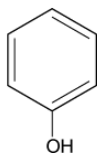


Figure 3 Percentage of apoptotic cells in the presence or absence of H_2O_2 and GAH. Error bars represent SEM.

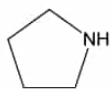
The side chain of the C-terminal amino acid residue in GAH is derived from which structure?

- ☐ A.  [10%]
- ☒ B.  [84%]
- ☐ C. 



[2%]

☐ D.

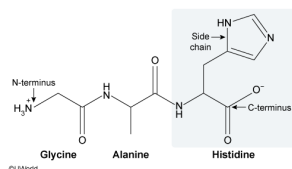


[2%]

Correct
84% answered correctly

Explanation:

Structure of GAH



Amino acids consist of both a **backbone structure** and a **side chain**. The backbones of all biologically relevant amino acids contain an **amino group** (N-terminus), an **α -carbon**, and a **carboxyl group** (C-terminus). The side chain is bonded to the α -carbon, and each of the 20 **standard amino acids** has a unique side chain structure, which determines the identity of the amino acid.

Peptides form when the C-terminus of one amino acid or peptide **reacts** with the N-terminus of another to form an amide bond. The newly formed peptide will have **one N-terminus** and **one C-terminus**. The amino acid residue closest to the N-terminus is the N-terminal residue, and the residue closest to the C-terminus is the C-terminal residue. Additional residues may be found between the N- and C-terminal residues.

The tripeptide GAH consists of an N-terminal glycine, an alanine, and a C-terminal histidine. The side chain of histidine is derived from the molecule imidazole, a **five-member aromatic ring** that contains **two nitrogen atoms** separated by one carbon atom. Therefore, the side chain of the C-terminal residue in GAH is derived from **imidazole**, shown in **Choice B**.

(Choice A) This is the structure of indole and is part of the side chain of **tryptophan** rather than histidine.

(Choice C) This structure is phenol. It is part of the side chain of **tyrosine**, not histidine.

(Choice D) This molecule is a cyclic secondary amine and is part of the structure of **proline**, not histidine.

Educational objective:

Amino acids consist of a side chain connected to a backbone structure that contains an amino group (N-terminus), an α -carbon, and a carboxyl group (C-terminus). Peptides are chains of amino acids that form when the amino group of an amino acid or peptide reacts with the carboxyl group of another amino acid or peptide. Like amino acids, peptides have an N-terminus and a C-terminus.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402726
System:
Amino Acids and Proteins

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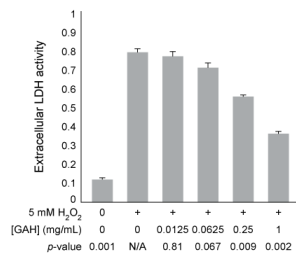


Figure 1 Extracellular LDH activity from PC12 cells exposed to H_2O_2 and varying concentrations of GAH. p -values are relative to 5 mM H_2O_2 and 0 mg/mL GAH.

Experiment 2

LDH was purified and assayed for activity in the presence of 5 mM H_2O_2 and varying concentrations of GAH (Figure 2).

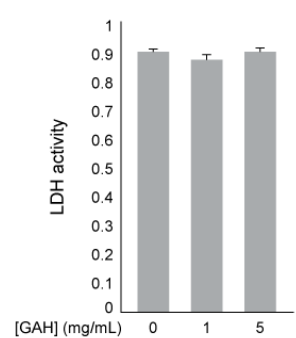


Figure 2 Activity of purified LDH with varying concentrations of GAH

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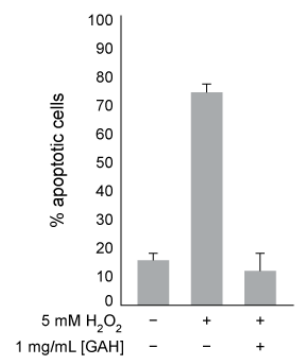


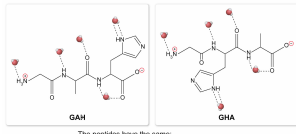
Figure 3 Percentage of apoptotic cells in the presence or absence of H_2O_2 and GAH. Error bars represent SEM.

The tested tripeptide variants most likely have which of the following characteristics in common?

- I. Primary structure
 - II. Isoelectric point
 - III. Hydrophobicity
- ☐ A. I and II only [13%]
- ☐ B. I and III only [12%]
- ☒ C. II and III only [62%]
- ☐ D. I, II, and III [11%]

Correct
62% answered correctly

Explanation:



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Peptides consist of two or more **amino acid residues** linked together by **peptide bonds**. The physical and chemical properties of peptides are determined by the identities of the amino acid residues and the sequence in which those residues are arranged (**primary structure**).

The **isoelectric point** of a peptide is the pH at which its **net charge is zero**. The net charge of a peptide at any given pH depends on the protonation state of its N- and C-termini, as well as on the protonation states of any ionizable side chains. These protonation states depend on the pK_a values (pH at which 50% of molecules are protonated) of the functional groups. In small peptides, the pK_a values of both termini and the side chains generally remain constant regardless of the sequence of residues. Therefore, all possible arrangements of the tripeptide have approximately the same net charge at any given pH, giving them the **same isoelectric point (Number II)**.

Hydrophobicity is a measure of a molecule's tendency (or lack thereof) to avoid **interactions with water** and is determined by the functional groups of a molecule. All tripeptides tested consist of the same three amino acids (A, G, and H), so they contain the same functional groups. Therefore, they most likely have the same level of hydrophobicity (ie, none of the tested peptides are very hydrophobic) (**Number III**).

(Number I) Primary structure is defined as the *sequence* of amino acids in a polypeptide chain. The tripeptides tested are all made of the same three amino acids but arranged in different orders, so the tripeptides do *not* all have the same primary structure.

Educational objective:

Peptides are chains of amino acids linked together by peptide bonds. For short peptides that do not adopt higher levels of structure, properties such as isoelectric point and hydrophobicity generally depend only on which amino acids are present (composition) and not on the order in which the amino acids are arranged (primary structure).

Time spent: 0 sec	QID: 402727
Subject: Biochemistry	System: Amino Acids and Proteins

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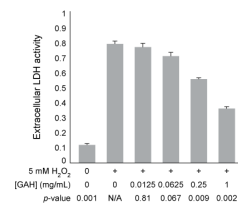


Figure 1 Extracellular LDH activity from PC12 cells exposed to H_2O_2 and varying concentrations of GAH. *p*-values are relative to 5 mM H_2O_2 and 0 mg/mL GAH.

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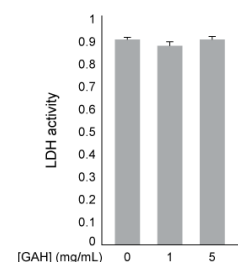


Figure 2 Activity of purified LDH with varying concentrations of GAH

Experiment 3

PC12 cells were incubated with or without 5 mM H_2O_2 in the presence and absence of 1 mg/mL GAH. Cells were then stained with the fluorescent marker TUNEL, which binds strongly to apoptotic cells. The percentage of apoptotic cells in each condition is shown in Figure 3.

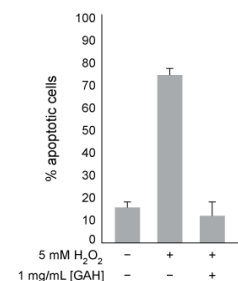


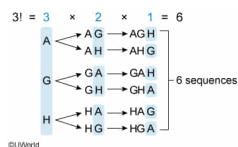
Figure 3 Percentage of apoptotic cells in the presence or absence of H_2O_2 and GAH. Error bars represent SEM.

Which of the following expressions gives the number of peptides that were tested for their ability to reduce extracellular LDH activity?

- ☒ A. $3!$ [63%]
☐ B. 3^3 [27%]
☐ C. 3^{20} [6%]
☐ D. 20^3 [2%]

Correct
 63% answered correctly

Explanation:



Peptides may be arranged in multiple ways, each with the constituent amino acids in different relative positions. For a peptide of known composition, in which each amino acid is different from every other amino acid, the number of possible arrangements is $n!$ (*n*-factorial), where n is the number of amino acids in the peptide.

For example, a dipeptide consisting of alanine and histidine can be arranged in $2! = 2$ different ways (AH or HA). A tripeptide made of three different, known amino acids can be arranged in $3! = 6$ different ways, a tetrapeptide can be arranged in $4! = 24$ different ways, and so on.

The passage states that researchers tested every possible sequence of tripeptides that contain one alanine, one glycine, and one histidine residue. In this case, $n = 3$, so **3! different combinations were tested**.

(Choice B) If alanine, glycine, and histidine could be present more than once per peptide (eg, GGG or HHA), there would be $3^3 = 27$ possible combinations.

(Choice C) If the peptide were 20 amino acids long instead of 3, and consisted of alanine, glycine, and histidine only, there would be 3^{20} possible combinations.

(Choice D) If all 20 standard amino acids were allowed in the tripeptides, instead of alanine, glycine, and histidine only, there would be 20^3 possible combinations.

Educational objective:

Peptides that consist of the same amino acids may be arranged in different ways. For a peptide consisting of n amino acids of known identity, in which no amino acids repeat, the number of possible combinations is $n!$.

Time spent: 0 sec
 Subject:
 Biochemistry

QID: 402728
 System:
 Amino Acids and Proteins

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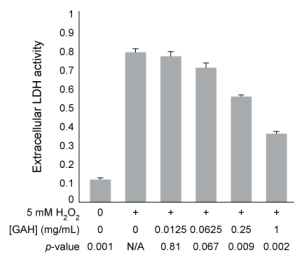


Figure 1 Extracellular LDH activity from PC12 cells exposed to H_2O_2 and varying concentrations of GAH. p -values are relative to 5 mM H_2O_2 and 0 mg/mL GAH.

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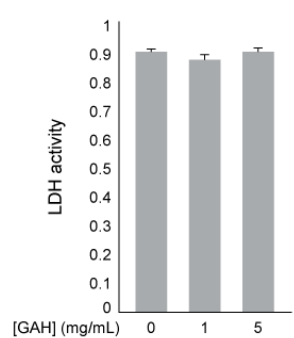


Figure 2 Activity of purified LDH with varying concentrations of GAH

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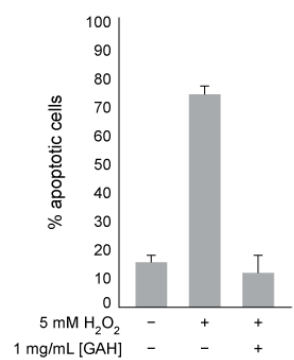


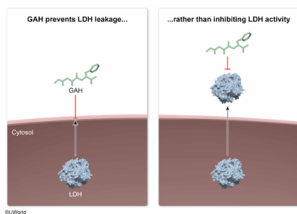
Figure 3 Percentage of apoptotic cells in the presence or absence of H_2O_2 and GAH. Error bars represent SEM.

The purpose of measuring purified LDH activity with various amounts of GAH was most likely:

- ☒ A. to demonstrate that GAH does not directly inhibit LDH. [45%]
- ☐ B. to determine baseline extracellular LDH activity. [18%]
- ☐ C. to show that GAH is an effective LDH antagonist. [31%]
- ☐ D. to characterize the GAH-LDH binding site. [4%]

Correct
45% answered correctly

Explanation:



sc000000

Peptides and proteins can interact with biological systems in a variety of ways, and the mechanisms by which they work must be **determined experimentally**. The results of an experiment often have **more than one interpretation**, and **additional experiments** must be performed to determine which interpretation is correct.

The passage states that LDH leakage is a marker of oxidative stress, and researchers measured extracellular LDH activity in the presence and absence of GAH. Experiment 1 shows that in the presence of GAH, extracellular LDH activity is reduced, indicating that GAH may decrease oxidative stress, which would in turn *decrease* LDH leakage.

However, it is also possible that GAH does *not* affect LDH leakage and that less LDH activity is observed because GAH directly inhibits LDH. To address this possibility, the researchers measured the activity of purified LDH (ie, no cells present) in the presence and absence of GAH in Experiment 2. They found that LDH activity was constant regardless of the amount of GAH used, and therefore they were able to demonstrate that **GAH does *not* directly inhibit LDH activity** but instead most likely prevents LDH leakage.

(Choice B) Baseline extracellular LDH activity was determined in Experiment 1 when cells were incubated in the absence of H_2O_2 and GAH. No cells were used in Experiment 2, so this experiment does *not* show baseline extracellular LDH activity.

(Choice C) **Antagonists** are molecules that bind to receptor proteins to inhibit a response. LDH is not a receptor protein, and there is no evidence presented in the passage that GAH binds to any receptor proteins.

(Choice D) Experiment 2 indicates that there are no direct interactions between GAH and LDH, so LDH most likely does not have a GAH-binding site.

Educational objective:

The mechanisms by which biological molecules interact must be determined experimentally. The results of an experiment may have multiple interpretations, and additional experiments must be performed to determine which interpretation is correct.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402729
System:
Amino Acids and Proteins

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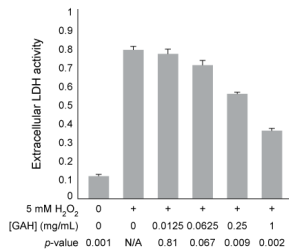


Figure 1 Extracellular LDH activity from PC12 cells exposed to H_2O_2 and varying concentrations of GAH. *p*-values are relative to 5 mM H_2O_2 and 0 mg/mL GAH.

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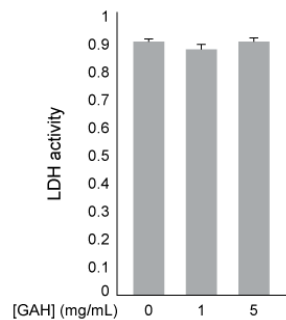


Figure 2 Activity of purified LDH with varying concentrations of GAH

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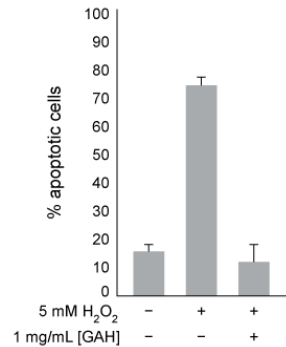


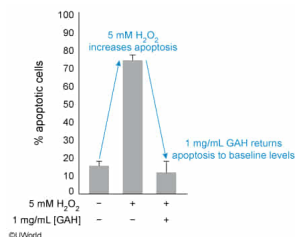
Figure 3 Percentage of apoptotic cells in the presence or absence of H_2O_2 and GAH. Error bars represent SEM.

The results of Experiment 3 best support which of the following statements?

- ☐ A. 5 mM H_2O_2 induced over 70% of all PC12 cells to undergo apoptosis. [16%]
- ☐ B. The effect of GAH on H_2O_2 -induced apoptosis is dose dependent. [1%]
- ☐ C. GAH decreases oxidative stress and apoptosis by reacting with H_2O_2 . [18%]
- ☒ D. 1 mg/mL of GAH brings apoptosis back to baseline level with 5 mM H_2O_2 present. [62%]

Correct
62% answered correctly

Explanation:



To determine the **effect of an independent variable** on a dependent variable, the condition in which all **independent variables are absent** should be tested initially. This condition establishes a **baseline** with which all other results can be compared.

Experiment 3 tested how apoptosis (dependent variable) was affected by H₂O₂ in the presence and absence of GAH (independent variables). To interpret the data, the results must be compared to levels of apoptosis when neither GAH nor H₂O₂ is present. Figure 3 shows that in the absence of both independent variables (control), approximately 15% of PC12 cells undergo apoptosis. When 5 mM H₂O₂ is added, the percentage of apoptotic cells increases to approximately 75%. When 1 mg/mL GAH is also present, the number of apoptotic cells is approximately 10%. This is within error of the percentage of apoptotic cells in the control. Therefore, **1 mg/mL of GAH brings apoptosis induced by 5 mM H₂O₂ back to baseline levels**.

(Choice A) Although approximately 75% of cells were apoptotic in the presence of 5 mM H₂O₂, Figure 3 shows that approximately 15% are *already* apoptotic without H₂O₂. Therefore, the H₂O₂ induced apoptosis in approximately 60% of cells, not over 70%.

(Choice B) A dose-dependent response would appear as a steady decrease in the percentage of apoptotic cells as the amount of GAH increases. Although it is likely that GAH decreases apoptosis in a dose-dependent manner, the results of Experiment 3 neither support nor refute this conclusion because only one concentration of GAH was tested.

(Choice C) Although it is true that GAH decreases oxidative stress and apoptosis, the results of Experiment 3 do not show whether or not GAH does so by *reacting* with H₂O₂. For example, it is possible that GAH reacts with something else that *subsequently* reacts with H₂O₂, or that GAH reverses damage induced by H₂O₂.

Educational objective:

Dependent variables are affected by independent variables. To accurately assess the effect of one or more independent variables on a dependent variable, a control in which all independent variables are absent must be performed. This control establishes a baseline to which all other results can be compared.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402730
System:
Amino Acids and Proteins

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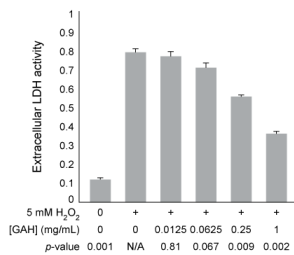


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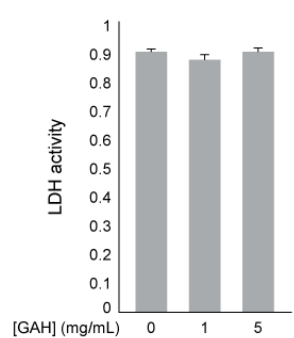


Figure 2 Activity of purified LDH with varying concentrations of GAH

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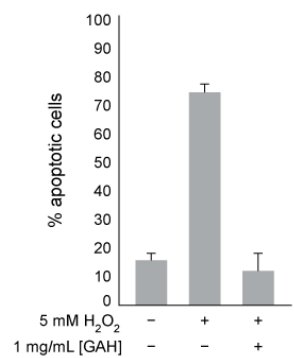


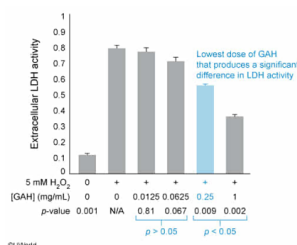
Figure 3 Percentage of apoptotic cells in the presence or absence of H_2O_2 and GAH. Error bars represent SEM.

What is the lowest tested concentration of GAH that produces a statistically significant reduction in extracellular LDH activity when 5 mM H_2O_2 is used?

- ☐ A. 0.0125 mg/mL [3%]
- ☐ B. 0.0625 mg/mL [28%]
- ☒ C. 0.25 mg/mL [56%]
- ☐ D. 1 mg/mL [12%]

Correct
56% answered correctly

Explanation:



p-values, expressed as decimals between 0 and 1, quantify the **probability** of observing a particular difference between two data sets due to **random chance** rather than two truly different results. For example, a *p*-value of 0.5 indicates that if an observed difference is merely due to random chance, one would expect to measure a similarly large difference in 50% of the tests performed, even if no changes were made relative to the control.

A **lower p-value** indicates a **lower probability of random chance**, and therefore a higher probability that the observed difference is real (ie, due to experimental differences). Generally, a difference between data sets is accepted as **statistically significant if $p < 0.05$** (ie, the experimental results would be observed less than 5% of the time due merely to random chance).

Figure 1 shows extracellular LDH activity for several concentrations of GAH, as well as their *p*-values relative to 5 mM H₂O₂ in the absence of GAH. Of the GAH concentrations tested, 0.25 mg/mL is the lowest concentration that yields a *p*-value of less than 0.05 ($p = 0.009$). Therefore, **0.25 mg/mL** is the lowest tested concentration that produces a significant reduction in extracellular LDH activity.

(Choice A) 0.0125 mg/mL of GAH yields nearly the same extracellular LDH activity as 0 mg/mL and produces a *p*-value of 0.81. For a difference to be statistically significant, the *p*-value must be less than 0.05.

(Choice B) Although the error bars between 0 mg/mL GAH and 0.0625 mg/mL GAH do not overlap, this would imply statistical significance only if the error bars represented a 95% confidence interval (as opposed to standard deviation or standard error of the mean). The passage does not state what **type of error bars** were used in this experiment. In contrast to error bars, *p*-values more reliably indicate significance or lack thereof. Because $p > 0.05$, 0.0625 mg/mL GAH does *not* produce a statistically significant decrease in LDH activity.

(Choice D) 1 mg/mL of GAH produces a significant reduction in oxidative stress ($p = 0.002$), but the question asks for the *lowest* tested dose that produces a significant response. 0.25 mg/mL *also* produces a significant reduction in oxidative stress, so 1 mg/mL is not the lowest dose that yields a significant response.

Educational objective:

p-values are statistical measures of the probability that an observed difference is due to random chance. Low *p*-values indicate a low probability that a difference between two observations is a coincidence. Generally, *p*-values of less than 0.05 (5% chance of coincidence) are accepted as statistically significant.

Time spent: 0 sec
 Subject:
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 Amino Acids and Proteins